

Book review

Open Innovation: The New Imperative for Creating and Profiting from Technology

Henry W. Chesbrough
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Chesbrough's central premise is that the apparent decline in the innovation capabilities of many leading companies at a time when so many promising and novel ideas pop up everywhere is due to the change in the way companies innovate. He suggests that we are witnessing a paradigm shift from "closed" to "open" innovation in how companies innovate and commercialize industrial knowledge. The main characteristic of the closed innovation model is vertically integrated research and development (R&D) departments that develop technology in-house for use in-house. Since inputs, outputs and processes remain within the boundaries of the firm, closure of the innovation process is achieved. In recent years, four factors challenged the closed innovation paradigm. First is the increasing amount of college and post-college training, resulting in a highly skilled workforce and large research laboratories. Next is the growing mobility of highly experienced and skilled people. Third is the mushrooming of private venture capital, which specializes in creating new firms to commercialize research. Finally, the reduced shelf life of technologies and increased competition with non-US firms eroded the effectiveness of the closed innovation paradigm.

Chesbrough subsequently concludes that a new paradigm, "open innovation", can better explain the innovation phenomena in many industries today. In open innovation, firms can and should use external ideas as well as internal ideas, and both internal and external paths to commercializing these ideas. The boundary of the firm becomes more porous, allowing greater mobility of ideas.

The book is organized into nine chapters. The first chapter illustrates the achievements and limitations of closed innovation by presenting Xerox's Palo Alto Research Center experience in the last two decades. In concluding the chapter, the author posits that venture capital (VC) has

changed how companies commercialize novel ideas. Chapter 2 examines the closed innovation model, concentrating on why it is obsolete. Chesbrough posits that the increasing availability and mobility of skilled workers, the opportunities that the VC market present, the availability of external options for novel ideas sitting on the shelves of the R&D departments and the increasing capability of external suppliers have all reduced the effectiveness of the closed innovation paradigm.

Chapter 3 discusses the open innovation paradigm in detail. Open innovation suggests that companies should make money by leveraging multiple paths to market for their technologies. R&D departments should not restrict their goal to exclusively inventing new knowledge. In addition, they should be open to accessing and integrating external knowledge. Finally, instead of managing intellectual property (IP) to exclude everyone else from using it, companies should manage IP to advance their own business models and to profit from others' use. One of the main business decisions of a firm becomes what necessary pieces of an innovation should be internally versus externally supplied, followed by how to integrate the pieces together.

In Chapter 4, Chesbrough examines the business model, which is the link between the innovation and the value a firm generates by commercializing the innovation. The author argues that the business model should:

- articulate the value proposition;
- identify a market segment;
- define the structure of the firm's value chain;
- specify and estimate the cost structure and target margins;
- describe the position of the firm within the supply chain value network; and
- formulate the competitive strategy.

Chesbrough ends Chapter 4 by demonstrating that the same technology taken to market through the two different business models will yield different amounts of value.

Chapters 5 through 7 explore different applications of open innovation at IBM, Intel and Lucent's New Ventures Group. The author indicates that IBM presents an excellent example of a firm that managed to transform its approach from closed to open innovation. While IBM has been successful with its internal research commitment to developing different business models for leveraging its technologies, Intel provides an example on the other end of the spectrum. It is a company that connects internal and external research with corporate VC to grow its business model.

After these three detailed case studies, the author turns to the issue of IP management in Chapter 8. Intelligent people with bright ideas are present inside every company as well as outside. The author claims that this poses substantial opportunities for firms; for example, it can improve the effectiveness of the innovation process by eliminating the wasteful duplication of innovative effort. However, ideas and knowledge are now flowing in and out of the company much more freely, which raises concerns about safeguarding IP. The author advocates companies being active buyers and sellers of IP. The chapter concludes with examples of IP management practices in several companies such as Millennium Pharmaceuticals, IBM and Intel.

The concluding chapter of the book, titled "Making the transition", explores the difficult task of moving from a closed innovation mentality to an open innovation mentality. The author suggests that a company should give up some amount of control in order to get access and utilize external ideas and knowledge.

The book is full of useful illustrations. Furthermore, in comparing closed and open innovation characteristics and different practices across firms, much of the information is illustrated by easy to access tables. On the other hand, the author indicates that open innovation applies to industries other than high-tech, but the chapters dedicated to illustrate open innovation contain companies only from high technology industries

such as Intel, IBM and Lucent. Consequently, the author's argument that the open innovation paradigm is applicable to various industries is not well supported.

Chesbrough targets both practitioners and scholars in presenting his views on the changing stage of innovation. For practitioners, especially if the locus of innovation is shifting in your business, then this book will be worth your while. Academics might be very much interested in some parts of the book. The author points towards universities where ideas could be obtained from faculty members. As the open innovation paradigm embraces external ideas and knowledge in conjunction with internal R&D, practitioners are advised to harness the novel ideas of scholars in order to come up with innovations that create value. The author indicates that sponsoring or supporting research is one way to learn about scholars' ideas. Consequently, if you are a scholar and are trying to persuade companies to support your research, this book is full of justifications as to why companies should invest in university research. Finally, *Open Innovation* would be a valuable supplementary book for an MBA course (especially for an executive MBA course) as it describes how the innovation process has evolved in recent years.

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